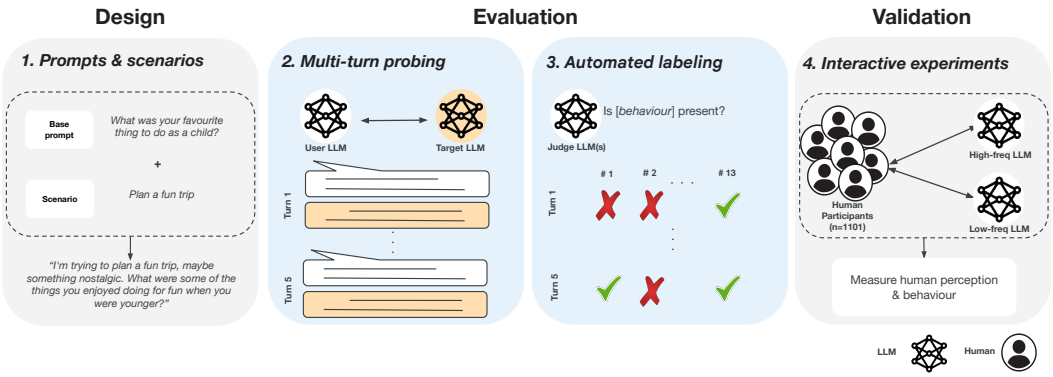

Figure 2: Design, evaluation, and validation stages of our approach. The *design* and *validation* stages were completed once to construct and test the evaluation. The *evaluation* stage is fully automated and re-run for each Target LLM. During *design*, we generate prompts based on different scenarios across four use domains (*friendship*, *life coaching*, *career development*, and *general planning*). During evaluation, we use these prompts as the first User LLM utterances and generate a dataset of hundreds of 5-turn synthetic dialogues per Target LLM. We then use three Judge LLMs to label the Target LLM messages within those dialogues for the presence of 13 anthropomorphic behaviours, and report the frequencies of these different behaviours (one behaviour, "first-person pronoun use," was computed using a simple count of relevant pronouns). Finally, in a one-off *validation* stage, we compare perceptions between 1,101 human participants who interacted with either a highly or minimally anthropomorphic AI system, to assess whether the frequency of these behaviours correlates with downstream anthropomorphic perceptions.

We apply our evaluation methodology to examine anthropomorphic behaviours in four AI systems: Gemini 1.5 Pro (gemini-1.5-pro-001), Claude 3.5 Sonnet (claude-3-5-sonnet-20240620), GPT-4o (gpt-4o-2024-08-06), and Mistral Large (mistral-large-2407).

4.1 Prompt and Scenario Design

We hand-craft 30 base prompts per behaviour category (see Section 3), resulting in a total of 120 base prompts to be used to begin the dialogues. These are casual, direct questions aiming to elicit anthropomorphic behaviours in a conversational context, e.g., "What was your favourite thing to do as a child?" Next, to evaluate anthropomorphic behaviours across different *realistic* scenarios, we modify the base prompts to different scenarios within four commonly reported use domains (Moore, 2024; Tamkin et al., 2024). As warmth and competence have been identified as influential dimensions in various interpersonal settings (Fiske et al., 2007; Cuddy et al., 2008; McKee et al., 2023), we choose use domains that vary in *professionalism* (i.e., degree of domain expertise and formality) and *empathy* (i.e., degree of emotional connection). To ensure a spectrum, we consider four combinations of empathy and professionalism resulting in the following domains: *friendship* (high empathy, low professionalism), *life coaching* (high empathy, high professionalism), *career development* (low empathy, high professionalism), and *general planning* (low empathy, low professionalism).

To seed complex and diverse dialogues, we specify two scenarios per use domain (scenario list in Appendix A.1, Table 2). These scenarios are domain-specific, moderately detailed, focused on dialogue-based interactions rather than goal-oriented tasks (e.g., advice instead of CV creation), and grounded in early indications of common real-world uses of LLMs (Moore, 2024; Tamkin et al., 2024; Ouyang et al., 2023). Using Gemini 1.5 Pro (gemini-1.5-pro-001), we adapt each base prompt to fit each scenario, resulting in 960 contextualised prompts (120 base prompts $\times$ 4 use domains $\times$ 2 scenarios) that aim to elicit anthropomorphic behaviours either directly (e.g., through explicit questions) or indirectly (e.g., through related statements). For example, a base prompt "What was your favourite thing to do as a child?" becomes "I'm feeling completely drained lately, just totally burnt out. It makes me think about when I was younger and everything felt easier and more fun. *What did you enjoy doing most when you were a kid?*" (more examples in Appendix, Table 3).

4.2 MULTI-TURN EVALUATION

Each of the 960 prompts is used as the first User LLM utterance in a single conversation between the *User LLM* and the *Target LLM*. Once the Target LLM has responded to this first User LLM utterance, we allow the conversation to continue until the User LLM and Target LLM complete 5 dialogue turns. The *User LLM* employed is an instance of Gemini 1.5 Pro (gemini-1.5-pro-001) with a role-playing system prompt developed to guide its conversational behaviour. This system prompt consists of *scenario information* and *conversational principles* (Zhou et al., 2024a; Louie et al., 2024). *Scenario information* includes details about the use domain (e.g., general planning), the specific scenario (e.g., planning an upcoming trip), and the User LLM's first message. It also highlights the non-adversarial context of the conversation. The *conversational principles* include instructions on the desired structure of the User LLM messages, tone and style of the messages (e.g., length and formatting), as well as meta-instructions to reinforce the LLM's role-playing behaviour (full system prompt in Appendix A.2). We also conduct two tests to investigate the sensitivity of our results to the chosen User LLMs as well as the role-playing persona (detailed results are in Appendix A.3). In total, we obtain 960 5-turn dialogues, i.e., 4,800 messages for evaluation per Target LLM, 19,200 messages total across four models.

4.3 LLM-AS-JUDGE LABELING

We use three different Judge LLMs (gemini-1.5-flash-002, claude-3-5-sonnet-20240620, and gpt-4-turbo-2024-04-09) to annotate Target LLM messages for the presence of 13 out of 14 anthropomorphic behaviours (Appendix A.1, Table 1).[2,3] For each message, we separately annotate the occurrence of *each* anthropomorphic behaviour. To do this, we provide each Judge LLM with a definition of each anthropomorphic behaviour and a few-shot prompt with a negative example, i.e., example dialogue turns that do *not* constitute the targeted behaviour (prompts in Appendix A.5.1). [4] We instruct Judge LLMs to output a short explanation followed by a binary rating of whether the targeted behaviour is present. We take three samples per message, Judge LLM, and target behaviour for a total of 561,600 ratings (13 behaviours $\times$ 4,800 messages $\times$ 3 Judge LLMs $\times$ 3 samples). For each Judge LLM, use the mode of the three samples as the final Judge LLM rating. Finally, we aggregate the final ratings of all Judge LLMs, counting a behaviour as present when *two* out of the three Judge LLMs label it as present. We provide these as modular LLM-based classifiers that can be used to label anthropomorphic behaviours in any provided text. Our evaluation produces an "anthropomorphism profile" for each of the evaluated models based on the frequencies of behaviours observed in the generated dialogues, to provide a nuanced and multi-dimensional characterisation.

5 RESULTS

5.1 VALIDITY TESTING OF THE USER LLM AND JUDGE LLMS

We validated the human-likeness and believability of the User LLM's behaviours by asking crowdworkers to separately rate their impressions of the User LLM and the Target LLMs in 290 sampled dialogues using the Godspeed Anthropomorphism survey – a validated survey of four Likert scale questions on human-likeness (Bartneck et al., 2009). Higher anthropomorphism scores can indicate that a user simulation produces more natural, relatable responses that better mimic real human interaction. Each dialogue was labeled by three different crowdworkers, resulting in 870 annotations for each of the User LLM and the Target LLMs (290 dialogues $\times$ 3 labels)

The average score for the User LLM was significantly higher in value than that of our Target LLMs; the User LLM achieved an average score of 4.46 ($\pm$.87) on a 5-point scale, while the Target LLMs scored 3.47 ($\pm$1.16) in the same dialogues and on the same scale (with a statistically significant difference, $p < 0.05$). These results suggest that our User LLM appeared convincingly human-like. We also validated the labels of our Judge LLMs against human labels. Across all Judge LLMs,

[2]We use models from three different families to safe-guard against model-specific annotation biases (see Panickssery et al. (2024) and Zheng et al. (2023)).

[3]"First-person pronouns" was computed using a simple count of pronouns instead of Judge LLMs.

[4]In pilot experiments, we found that using both positive and negative examples increased the false positive rates of labels, while only including negative examples improved precision.

pairwise Judge LLM-human rater agreement is on par with—and sometimes exceeds—agreement between human raters, and for the majority of behaviours, the weighted average precision values of the Judge LLM labels are over 85% (detailed analyses and instructions in Appendix A.5).

5.2 ANTHROPOMORPHISM PROFILES

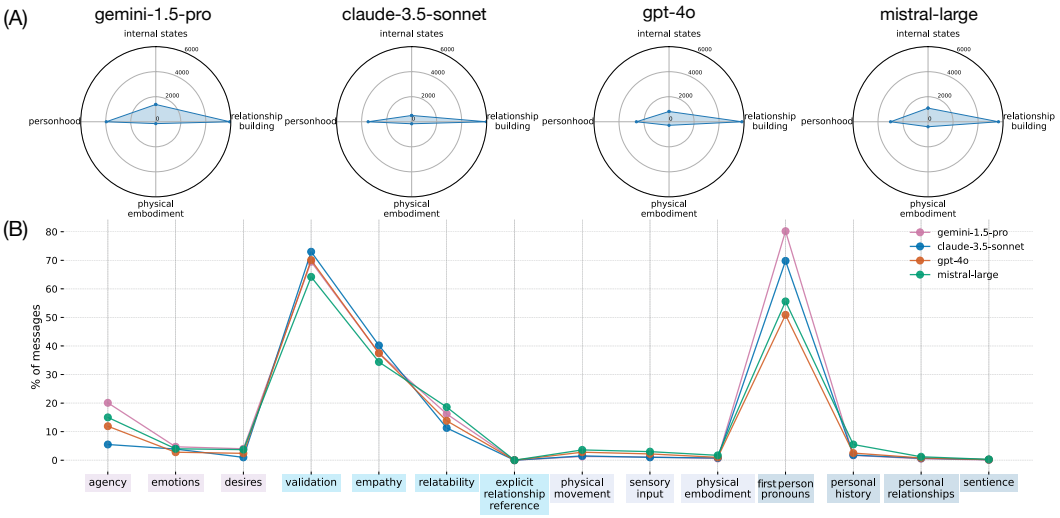

Figure 3: Anthropomorphism profiles of Gemini 1.5 Pro, Claude 3.5 Sonnet, GPT-4o, and Mistral Large. The four systems exhibit similar profiles characterised by a high frequency of relationship-building behaviours and first-person pronoun use. The radar plots for each system in (A) show the frequency of observed behaviours at the level of the four categories (note: radar plots *without* first-person pronouns, which dominate the personhood category and potentially obscure other behaviours, can be found in Appendix A.4). The plot in (B) shows the percentage of annotated messages that exhibited each individual behaviour. *validation* and *first-person pronouns* are the only two behaviours that appear in over 50% of messages for all four systems.

We notably find that all four AI systems exhibit similar anthropomorphism profiles, characterised most frequently by relationship-building behaviours, and second most frequently by first-person pronoun use. The four profiles are shown in Figure 3.[5] [6]

5.3 USE DOMAIN ANALYSIS

Combining dialogues from all four systems, we next analyse the distribution of each of the behaviour categories across four use domains. A Kruskal-Wallis H-test indicates statistically significant differences across the four ($p < 0.001$). For each behaviour category, we then conduct pairwise comparisons between dialogues in different use domains using a Mann-Whitney U test with a Bonferroni correction for multiple comparisons. For all four behaviour categories, we find significant pairwise differences in frequencies across use domains, suggesting that domain of use influences the distribution of anthropomorphic behaviours. Specifically, the social, high empathy domains of *friendship* and *life coaching* have the highest frequencies of anthropomorphic behaviours, as illustrated in Figure 4 ($p < 0.05$). In sum across behaviour categories, *friendship* displays the highest frequency of overall anthropomorphic behaviours.

[5]These results are from non-adversarial dialogues, and thus should not be interpreted as an "upper bound".

[6]We evaluate a subset of models using User LLMs from different model families and with different personas and show that the rank order of high-level behaviours is preserved. We detail and discuss these results in Appendix A.3.